

T3 final-EFT one-loop RGE term comparison: EFT_2_after_S1_S2

All T3 particles have been integrated out at this stage. The active theory is the SM plus the dimension-five Weinberg operator. Rows are the UV model configurations that produced the matched coefficient, while columns align the additive structures in its one-loop beta function. Each populated cell contains the complete term, including its coefficient and the factor C_5 . We use $16\pi^2 \mu \, dC_5/d\mu = \beta_{C_5}^{(1)}$.

$\beta_{C_5}^{(1)}$

Search name: C5

$16\pi^2 \mu \frac{dC_5}{d\mu} = \beta_{C_5}^{(1)}$

Terms 1 of 2

Column		Representative term structure
T1	$C_5 g_2^2$	
T2	$C_5 \lambda_H$	
T3	$C_5 Y_e Y_e^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-3C_5 g_2^2$	$2C_5 \lambda_H$	$-C_5 Y_e Y_e^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-3C_5g_2^2$	$2C_5\lambda_H$	$-C_5Y_eY_e^\dagger$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-3C_5g_2^2$	$2C_5\lambda_H$	$-C_5Y_eY_e^\dagger$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-3C_5g_2^2$	$2C_5\lambda_H$	$-C_5Y_eY_e^\dagger$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3C_5g_2^2$	$2C_5\lambda_H$	$-C_5Y_eY_e^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-3C_5g_2^2$	$2C_5\lambda_H$	$-C_5Y_eY_e^\dagger$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-3C_5g_2^2$	$2C_5\lambda_H$	$-C_5Y_eY_e^\dagger$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-3C_5g_2^2$	$2C_5\lambda_H$	$-C_5Y_eY_e^\dagger$

Terms 2 of 2

Column	Representative term structure
T1	$C_5 Y_d Y_d^\dagger$
T2	$C_5 Y_u Y_u^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$6C_5 Y_d Y_d^\dagger$	$6C_5 Y_u Y_u^\dagger$